

YOUNGSIK CHOI

Oak Ridge National Laboratory, Oak Ridge, TN, USA
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FIELDS OF INTEREST

- Building energy modeling and simulation
- Energy-efficient HVAC control and occupant comfort
- AI-driven methods for building design and operation

ACADEMIC APPOINTMENTS

Oak Ridge National Laboratory
Postdoctoral Research Associate

Oak Ridge, TN
Jul. 2026 – Present

Texas A&M University
Research Associate

College Station, TX
May 2026 – Jul. 2026

EDUCATION

Texas A&M University

College Station, TX

Ph.D. in Mechanical Engineering

May 2026

***Dissertation:** Optimizing Dedicated Outdoor Air Systems and Exploring Synergies with Heat Pumps for Energy-Efficient Buildings*

***Advisor:** Prof. Zheng O'Neill, PhD, PE, FASHRAE, FIBPSA*

***GPA:** 3.84/4.0*

Seoul National University

Seoul, South Korea

M.S. in Architectural Engineering (Built Environment Track)

Feb. 2022

***Thesis:** Stochastic Setpoint Temperature Learning for Occupant Behavior-based Control*

***Advisor:** Prof. Cheol-Soo Park, PhD, FIBPSA*

***GPA:** 3.97/4.3*

Seoul National University

Seoul, South Korea

B.S. in Architectural Engineering (Built Environment Track)

Feb. 2020

Graduated *cum laude*

***GPA:** 3.88/4.3*

Peking University

Beijing, China

Undergraduate Exchange Student

Jul. 2016

Participated in the 2016 Globex Julmester Program.

RESEARCH EXPERIENCE

Postdoctoral Research Associate

Jul. 2026 – Present

Integrated Building Deployment and Analysis Group, Oak Ridge National Laboratory

- Contribute to U.S. DOE projects on building energy modeling, **retrofit analysis**, and **AI integration** into building energy simulation.
- Support the DOE Weatherization Assistance Program (WAP) in developing **retrofit priority lists** for multifamily buildings serving low-income households.

Graduate Research Assistant

May 2022 – May 2026

Building Energy and HVAC&R Research Group, Texas A&M University

Advisor: Dr. Zheng O'Neill

- Impact of Residential Mid-Efficiency Inverter AC/HPs on Texas Electric Grid

Funded by Daikin Comfort

Feb. 2026 – Jul. 2026

Roles: Lead Graduate Researcher (Feb. – May 2026); Research Associate (Jun. – Jul. 2026)

- **Calibrated** the TAMU Smart Home virtual testbed model to simulate zone air temperature and air-source heat pump performance in heating and cooling modes.
- Developed a *ResStock*-based **building stock modeling** framework to assess HVAC replacement scenarios and the deployment potential of variable-speed air-source heat pumps across Texas.

- Smart Meter Data Analysis

Funded by Texas A&M University

Jun. 2024 – May 2026

Roles: Graduate Researcher

- Contributed to analyses of **smart meter electricity data** from more than 1.93 million residential buildings in Harris County, Texas, focusing on demand flexibility and energy resilience. [\[J4\]](#)

- High-performance Whole Building Design 3D-printed Carbon-Absorbing Funicular Structures

Funded by *ARPA-E HESTIA* Program: Lead Graduate Researcher

Jan. 2023 – Jun. 2025

Roles: Lead Graduate Researcher

- Developed models of **radiant systems coupled** with **ground-source heat pumps** in buildings with carbon-absorbing funicular structures using EnergyPlus and GLHEPro.
- Demonstrated the effectiveness of **mixed-mode ventilation** integrating natural ventilation with a dedicated outdoor air system.
- Analyzed **long-term operational energy use and carbon emissions** using future weather files (*fTMY*) and emissions factors from the NREL *Cambium* dataset.
- Presented research at the *2024 ARPA-E Energy Innovation Summit*.

- Optimizing Supply Air Temperature Control for Dedicated Outdoor Air Systems

Funded by ASHRAE RP-1865

May 2022 – Mar. 2025

Roles: Lead Graduate Researcher

- Developed *EnergyPlus* models of **dedicated outdoor air systems (DOAS)** coupled in parallel with heat pumps, fan coil units, and chilled beams. [C1,5]
 - Optimized **DOAS operating strategies** using *genetic algorithms* on Texas A&M's **High Performance Research Computing** (HPRC) system. [J5]
 - Designed a *machine learning*-based, **optimization-informed supply air temperature reset strategy** for DOAS to support updates to ASHRAE Guideline 36. [J2]
 - Presented research progress at six ASHRAE meetings.
 - **Collaborated with industry partners** to incorporate practical insights into the research.
- Proposal Development
 - *Impact of Residential Mid-Efficiency Inverter AC/HPs on Texas Electric Grid* funded by Daikin
 - Contributed to proposal drafting.
 - *Low-Cost, Durable, and Resistant Radiative Cooling Roof for Enhanced Energy Efficiency of Low- and Medium-Income Houses* funded by DOE
 - Performed energy simulations to assess radiative cooling roof applications in residential buildings.

ORISE Graduate Intern

Jun. 2025 – Aug. 2025

Oak Ridge National Laboratory

Mentors: Drs. Piljae Im and Yeobeom Yoon

- Development and Calibration of Building Energy Models
 - Developed a **building energy model** for a city-owned commercial building using architectural and mechanical drawings. Created building geometry in *SketchUp* with the *Euclid* plug-in and implemented HVAC systems in *EnergyPlus*. Calibrated the model using measured hourly electricity and gas consumption data.
- Fault Detection and Diagnosis of Refrigerant Charge Faults in Residential HVAC Systems
 - Contributed to **automated fault detection and diagnosis (AFDD)** models for refrigerant charge faults in residential HVAC systems using real-world testbed data.
 - Conducted a systematic literature review on AFDD in residential buildings.

Ph.D. Intern

Jul. 2023 – Aug. 2023

Pacific Northwest National Laboratory

Mentors: Drs. Yan Chen and Xuechen (Jerry) Lei

- Reviewed construction weights and bottom-up building energy modeling methods for the **construction weight analysis** project.

Graduate Research Assistant

Jan. 2020 – Feb. 2022

Building Simulation Lab, Seoul National University

Advisor: Dr. Cheol-Soo Park

- Development of Real-time Diagnosis Technology of Home Energy Usage and Smart & Autonomous Control/Management System

Funded by Korean Energy Technology Evaluation and Planning (KETEP)

Jan. 2020 – Feb. 2022

Roles: Graduate Researcher

- Supported *EnergyPlus* model development and **sensitivity analyses** to identify key input parameters before sensor installation. [C13-15]
 - Performed **lighting simulations** in Relux to size and select dimmable lighting for residential apartments.
 - Installed **Internet of Things (IoT) sensors** and collected data on indoor environmental conditions, occupant behavior, and energy use in existing residential buildings. Supported real-time data queries using *Python*.
- Development of Building Energy Management System (BEMS) Energy Saving Algorithms

Funded by Hyundai Development Company (HDC) I-Controls

Jun. 2020 – Feb. 2021

Roles: Graduate Researcher

- Developed a *machine learning* model to predict peak electricity demand for a building energy management system (BEMS) in a high-rise office building using measured data.
- Investigated virtual-to-real *transfer learning* for indoor air temperature prediction using *EnergyPlus* simulation outputs and real-world data. [C11]

TEACHING EXPERIENCE

Teaching Support, Texas A&M University

Fall 2023

- TAMU MEEN 489: ASHRAE 2024 Design Competition
- Mentored an undergraduate capstone team participating in the *ASHRAE 2024 Design Competition* and provided technical guidance on building energy modeling with *SketchUp* and *EnergyPlus*.

Graduate Teaching Assistant, Seoul National University

Fall 2020

- Creative Engineering Design
- Supported instruction on IoT applications in architecture, including *Arduino* programming and environmental sensors.
- Guided students in *Arduino* programming and sensor implementation during labs and projects; assisted with grading.

Peer Tutor, Seoul National University via *Work-Study Scholarship*

Fall 2019

- Selected on the basis of academic performance to provide one-on-one tutoring in *Fluid Systems in Buildings* and *Design and Construction of Steel Structures*.

Related Program

Academy for Future Faculty, Texas A&M University

Spring 2026

- Earned *CIRTL Associate* status through a teaching development program for graduate students and postdoctoral scholars.

PEER-REVIEWED JOURNAL PAPERS

- [J1] Choi, Y. J., Yoon, Y., Im, P., Jung, S., Kolar, B., Dyer, T., & **Choi, Y.** (2026). Field-based AFDD for refrigerant undercharge in residential HVAC systems: enhancing reliability through false alarm mitigation. *Energy and Buildings*, 117582.
- [J2] **Choi, Y.**, O'Neill, Z., & Zhou, X. (2026). Optimization-informed heuristic rules for supply air temperature reset in dedicated outdoor air systems. *Energy and Buildings*, 117199.
- [J3] **Choi, Y.**, Lu, X., & O'Neill, Z. (2026). Adopting Ground Source Heat Pumps in Commercial Buildings: Nationwide Analysis of Energy-saving and Decarbonization Potentials. *Energy Conversion and Management*, 348, 120635.
- [J4] Guo, M., **Choi, Y.**, Cheong, S. M., & O'Neill, Z. (2025). Current and Future Residential Electricity Demand Using Large-Scale Smart Meter Data in a Changing Climate. *Sustainable Cities and Society*, 106623.
- [J5] **Choi, Y.**, Lu, X., Feng, F., & O'Neill, Z. (2024). Large-scale energy cost optimization and performance analysis for dedicated outdoor air system: simulation results from ASHRAE RP-1865. *Science and Technology for the Built Environment*, 30(10), 1217-1235.
- [J6] **Choi, Y.**, Lu, X., O'Neill, Z., Feng, F., & Yang, T. (2023). Optimization-informed rule extraction for HVAC system: A case study of dedicated outdoor air system control in a mixed-humid climate zone. *Energy and Buildings*, 113295.

CONFERENCE PAPERS

- [C1] **Choi, Y.**, O'Neill, Z., & Zhou, X. (2025). Optimization-informed Dedicated Outdoor Air System Supply Air Temperature Reset Strategy. In Proceedings of *2025 ASHRAE Annual Conference*.
- [C2] **Choi, Y.**, Lu, X., Feng, F., & O'Neill, Z. (2024). Energy Saving Potential Analysis for Primary Schools with Optimal Dedicated Outdoor Air System Control in Different Climate Zones. In Proceedings of *2024 ASHRAE Winter Conference*.
- [C3] **Choi, Y.**, Lu, X., O'Neill, Z., & Feng, F. (2023). Optimal Supply Air Temperature Control for Dedicated Outdoor Air System Under Varying Climate Zones. In Proceedings of *Building Simulation Conference 2023*.
- [C4] **Choi, Y.**, O'Neill, Z., & Yang, S. (2023). Potentials of Direct Air Capture (DAC) of CO₂ in a Dedicated Outside Air System (DOAS). In Proceedings of *2023 ASHRAE Annual Conference*.
- [C5] **Choi, Y.**, Lu, X., O'Neill, Z., & Pang, Z. (2023). Modeling and Simulation of Dedicated Outdoor Air System (DOAS) with a Passive Desiccant Wheel: A Case Study using EnergyPlus. In Proceedings of *2023 ASHRAE Annual Conference*.

- [C6] **Choi, Y.**, Shin, H. S., Cho, S., Ko, Y. D., & Park, C. S. (2020). Predictive Uncertainty of Residential Building Energy Model. In Proceedings of 2020 Winter Simulation Conference. (**Best Poster Award Runner-up**)
- [C7] Cho, S., Shin, H. S., **Choi, Y.**, Ko, Y. D., & Park, C. S. (2020). Occupant-adaptive indoor environmental controller using DQN. In Proceedings of 2020 Winter Simulation Conference.
- [C8] Cho, S., **Choi, Y.**, Kim, J. H., & Park, C. S. (2022). Integrated control of radiant floor heating system in residential buildings. In Proceedings of 2022 Spring Annual Conference of the Architectural Institute of Korea [In Korean] (**Outstanding Paper Award**)
- [C9] Cho, S., Kim, J. H., **Choi, Y.**, & Park, C. S. (2022). Uncertainty analysis of cooling energy in residential building. In Proceedings of 2022 Spring Annual Conference of the Architectural Institute of Korea [In Korean]
- [C10] Ko, Y. D., Shin, H., Cho, S., **Choi, Y.**, & Park, C. S. (2021). Identification of occupant behavior patterns from aggregated household energy data using deep learning. In Proceedings of 2021 Winter Conference of the Korean Institute of Communications and Information Sciences [In Korean]
- [C11] **Choi, Y.**, Yi, D. H., Shin, H., Chu, H. G., Yoo, S., & Park, C. S. (2020). Application of transfer learning to a simulation model for room air temperature. In Proceedings of 2020 Autumn Annual Conference of the Architectural Institute of Korea [In Korean]
- [C12] **Choi, Y.**, Shin, H., Ko, Y., Cho, S., & Park, C. S. (2020). Predictive uncertainty of energy simulation model using Deep Ensembles. In Proceedings of 2020 Spring Annual Conference of the Architectural Institute of Korea, Vol. 40-1, pp. 290-291 [In Korean]
- [C13] Ko, Y. D., Shin, H., Cho, S., Choi, Y., & Park, C. S. (2020). Uncertainty in dynamic sensitivity analysis of occupant behavior for cooling energy in residential buildings. In Proceedings of 2020 Spring Annual Conference of the Architectural Institute of Korea [In Korean]
- [C14] Cho, S., Shin, H. S., Ko, Y. D., **Choi, Y.**, & Park, C. S. (2020). Static vs. dynamic sensitivity analysis of occupant behavior for energy consumption of cooling season in residential buildings. In Proceedings of 2020 Spring Annual Conference of the Architectural Institute of Korea [In Korean]
- [C15] Shin, H., Ko, Y. D., Cho, S., **Choi, Y.**, & Park, C. S. (2020). Quantifying the relevance of occupant behavior to building energy use. In Proceedings of 2020 Spring Annual Conference of the Architectural Institute of Korea [In Korean]

TECHNICAL REPORTS

- [TR1] ASHRAE Research Project Report 1865-RP: Optimizing Supply Air Temperature Control for Dedicated Outdoor Air System.

PRESENTATIONS AND TALKS

1. “ASHRAE RP-1865: Optimization of Supply Air Temperature for DOAS”, **Invited talk** at 2026 ASHRAE Winter Conference, Las Vegas, NV, USA, Feb. 3, 2026.
2. “Model Predictive Control-Informed Rule-Based Strategy for Enhancing Pre-Cooling Operations in Residential Buildings during Heat Waves: Simulation and Field Testing”, **Seminar talk** at 2025 ASHRAE Annual Conference, Phoenix, AZ, USA, Jun. 25, 2025.

3. “Optimization-informed Dedicated Outdoor Air System Supply Air Temperature Reset Strategy”, **Oral presentation** at *2025 ASHRAE Annual Conference*, Phoenix, AZ, USA, Jun. 25, 2025.
4. “Modeling and Control of Smart HVAC Systems: Insights from Research on Dedicated Outdoor Air Systems”, **Invited talk** at Pukyong National University, Busan, South Korea (Virtual Seminar), Apr. 29, 2025.
5. “Energy Saving Potential Analysis for Primary Schools with Optimal Dedicated Outdoor Air System Control in Different Climate Zones”, **Oral presentation** at *2024 ASHRAE Winter Conference*, Chicago, IL, USA, Jan. 21, 2024.
6. “Model predictive control of radiant heating system under varying thermal mass scenarios in a mixed-humid climate zone”, **Oral presentation** at *2024 Texas A&M Conference on Energy*, College Station, TX, USA, Sep. 13, 2024.
7. “Large-scale Energy Performance Analysis for Optimal Control for Dedicated Outdoor Air System”, **Poster presentation** at *the 1st International Workshop on Building and Simulation (BAS 2024)*, Syracuse, NY, USA, May 13, 2024.
8. “A Collage of ASHRAE Research Projects at Building Energy & HVAC Research Group at the Texas A&M University”, **Invited talk** at *Austin ASHRAE 2024 Expo*, Norris Conference Center - Austin, Austin, TX, Apr. 4, 2024.
9. “Optimal Supply Air Temperature Control for Dedicated Outdoor Air System Under Varying Climate Zones”, **Oral presentation** at *Building Simulation Conference 2023*, Shanghai, China, Sep. 4, 2023.
10. “Modeling and Simulation of Dedicated Outdoor Air System (DOAS) with a Passive Desiccant Wheel: A Case Study using EnergyPlus”, **Oral presentation** at *2023 ASHRAE Annual Conference*, Tampa, FL, USA, Jun. 25, 2023.
11. “Potentials of Direct Air Capture (DAC) of CO₂ in a Dedicated Outside Air System (DOAS)”, **Poster presentation** at *2023 ASHRAE Annual Conference*, Tampa, FL, USA, Jun. 25, 2023.
12. “Predictive Uncertainty of Residential Building Energy Model”. **Poster presentation** at *2020 Winter Simulation Conference*, Orlando, FL, USA (Virtual Conference), Dec. 14–18, 2020. (**Best Poster Award Runner-up**)
13. “Application of transfer learning to a simulation model for room air temperature”, **Oral presentation** at *2020 Autumn Annual Conference of the Architectural Institute of Korea*, Yeosu, South Korea (Virtual Conference), Oct. 26, 2020.
14. “Predictive uncertainty of energy simulation model using Deep Ensembles”, **Oral presentation** at *2020 Spring Annual Conference of the Architectural Institute of Korea*, Gyeongju, South Korea (Virtual Conference), Apr. 24, 2020.

AWARDS, GRANTS, AND FELLOWSHIPS

Melbern G. Glasscock '59 Endowed Graduate Fellowship in Mechanical Engineering	2025
J. Mike Walker '66 Department of Mechanical Engineering, Texas A&M University; 3,000 USD	
Brenda & Jerry Gray '62 Fellowship	2024
J. Mike Walker '66 Department of Mechanical Engineering, Texas A&M University; 1,000 USD	

ASHRAE Graduate Grant-In-Aid Scholarship American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE); 10,000 USD	2024
Graduate Student Research and Presentation Travel Award Graduate and Professional Studies, Texas A&M University; 750 USD	2023
Graduate Student Travel Award J. Mike Walker '66 Department of Mechanical Engineering, Texas A&M University; totaling 2,250 USD	2023, 2024, 2025
An AI for IoT Information (AI3) Prize Competition Phase 1 winner (the only student team); \$10,000 team award.	2023
Emil Buehler Aerodynamic Analog Fellowship J. Mike Walker '66 Department of Mechanical Engineering, Texas A&M University; 3,000 USD	2022
Outstanding Paper Award <i>The 2022 Spring Annual Conference of the Architectural Institute of Korea</i>	2022
Best Poster Award (runner-up) <i>The 2020 Winter Simulation Conference</i>	2020
Certificate of Appreciation <i>The 2018 Key Resolve R.O.K & U.S. Joint Exercise</i>	2018
Merit-based Scholarships before 2022 (approximately \$20,000 total equivalent)	
• Academic Eminence Scholarships funded by Seoul National University (full tuition; Spring 2016, Fall 2018, Spring 2019, Fall 2019, partial tuition; Spring 2015) • Organization Scholarships funded by the Education and Research Foundation of Seoul National University (full tuition for 3 semesters; Fall 2020, Spring 2021, Fall 2021), Brain Korea 21 (monthly stipend; Fall 2020), Moon-Ju Scholarship Foundation (full tuition for 1 semester; Fall 2015) • Work-Study Scholarships funded by Seoul National University (monthly stipend; Fall 2014, Spring 2015, Fall 2019)	

SERVICE AND LEADERSHIP

Professional Memberships

- The American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE)
 - Student member (2023 – Present)
 - Texas A&M Student Chapter member (2023 – 2026)
- Architectural Institute of Korea (AIK)
 - Student member (2020–2021)

Academic Committee

- ASHRAE Technical Committee (TC) activities
 - Corresponding member, TC 7.6 (Building Energy Performance) (2023 – Present)
 - Provisional corresponding member, TC 7.5 (Smart Building Systems) (2023 – Present)
 - Provisional corresponding member, TC 8.10 (Mechanical and Desiccant Dehumidification Equipment, Heat Pipes and Components) (2025 – Present)

Reviewer

- *Journal of Building Performance Simulation* [IBPSA's official journal] (2025 – Present)
- *Science and Technology for the Built Environment* [ASHRAE's official journal] (2026 – Present)
- *ASME Journal of Engineering for Sustainable Buildings and Cities* (2025 – Present)
- *Journal of Building Engineering* (2025 – Present)
- *Applied Thermal Engineering* (2026 – Present)
- *Springer Nature Scientific Reports* (2026 - Present)
- ASHRAE Handbook – HVAC Applications, Chapter 43: “Supervisory Control Strategies and Optimization” (2025 – 2026)

Mentorship

- Mentored undergraduate and graduate students at Texas A&M University. (2022 – 2026)
- Peer Counselor (2018)
 - Served as a trained peer counselor for approximately 50 personnel during mandatory military service in the R.O.K. Naval Mobile Construction Squadron.
- Mentor via *Seoul National University Dream Consultant program* (2014)
 - Developed and delivered a mentoring program for high school students from underserved communities.

Seoul National University Buddy Assistants

- Team Leader (2015)
 - Led approximately 20 team members in organizing social programs for about 400 international exchange students at Seoul National University, achieving the highest end-of-term program evaluation scores.

REFERENCES

Available upon request.